

FX Fundamentals

What is FX Trading?

- Bet on the direction of an exchange rate
 - One currency over another – WHICH will do well
- The largest financial market by volume, it operates 24/7
- Highly macro-relevant:
 - Interest rates / monetary policy
 - Geopolitical events
 - Economic data: GDP growth, inflation, unemployment rates
 - Liquidity

Exchange rate regimes

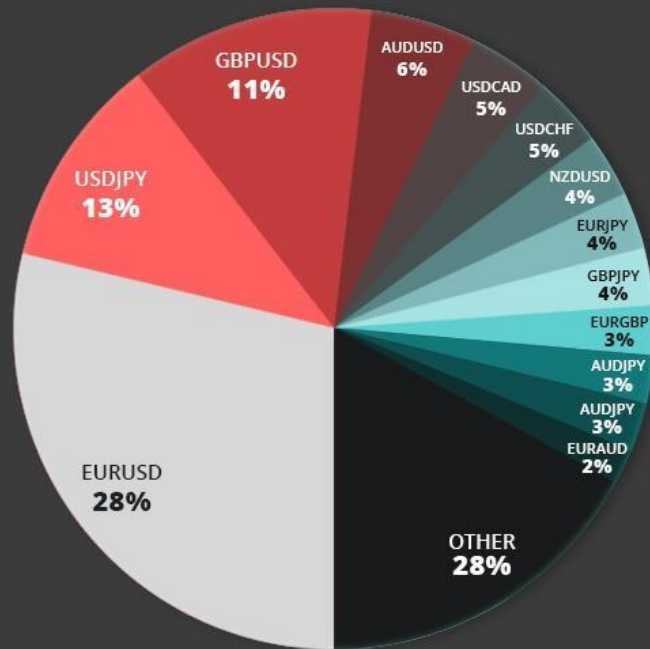
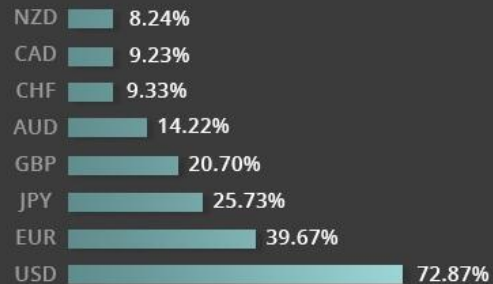
- Different economies have different exchange rate regimes
 - Floating: most countries in the world (liquid, free)
 - Fixed: countries that want better economic credibility / stability e.g. USD/KZT (Kazakhstan)
 - Managed floating: countries that want both e.g. China
- Fixed / managed floating regimes are traded via NDFs (not your typical FX spot / forwards)
 - Because you can't easily get capital out of the country – there is not much KZT outside Kazakhstan! So, you can only settle your USD/KZT trades in dollars only

G10 vs. EM

- G10: the 10 most liquid currencies, all advanced economies (except China)
- EM: less liquid currencies, in developing economies

Most liquid currency pairs

By currencies



Liquidity

- Liquidity: the extent to which an asset can be bought / sold quickly at stable prices
- Liquidity can be assessed by bid/ask spread
 - Bid price: the price at which something is bought
 - Ask price: the price at which something is sold
 - Market makers (the big banks) make their profits based on this spread
- If a market has high liquidity e.g. US Treasuries or EURUSD, the bid-ask spread will be low, as the order book is full of supply and demand
- If a market has low liquidity, such as Sri Lankan Government Bonds or some distressed corporate bond, spreads will be wider (i.e you may not be able to buy/sell them at their desired price)
- Liquidity can fall quickly in market crises

Role of CBs

Intervention mechanisms:

- 1. Interest rates
- 2. Balance sheet management i.e. buying / selling government bonds
- 2. Forward guidance
- 3. Direct FX intervention

FX instruments

- Spot is affected by the following factors:
 1. Relative economic health (e.g. Current Account, terms of trade)
 2. Reserves (intervention)
 3. Monetary policy spot usually strengthens in rate hike env.
 4. Geopolitical risk
- Forwards determined by interest rate parity & relative funding premium of one currency to another

Spot

- FX spot rate is the value of base currency in relation to term currency
- For USD/EUR, USD is the base, EUR is the term
- If USD/EUR is quoted at 0.92, it means \$1 for 0.92 EUR
- P&L are reported in the denominator currency
- Spot rate refers to settlement in T+2 (2 business day's time) with exceptions such as USDCAD where it is T+1

	Dates	Pts Bid	Pts Ask	Fwds Bid	Fwds Ask
T	10/19/22	-5.26	-4.24	7.208877	7.210157
1N	10/20/22	-6.31	-1.53	7.208453	7.209631
1P	10/20/22	7.2083	7.2090	7.2083	7.2090
1N	10/21/22	-3.98	-3.02	7.207902	7.208698
1W	10/27/22	-26.54	-16.36	7.205646	7.207364
2W	11/03/22	-45.71	-38.53	7.203729	7.205147
3W	11/10/22	-71.16	-61.58	7.201184	7.202842
1M	11/21/22	-114.02	-101.98	7.196898	7.198802
2M	12/20/22	-228.16	-210.84	7.185484	7.187916
3M	01/20/23	-379.14	-359.86	7.170386	7.173014
4M	02/21/23	-537.90	-520.14	7.154510	7.156986
5M	03/20/23	-674.06	-653.10	7.140894	7.143690
6M	04/20/23	-822.92	-803.08	7.126008	7.128692
9M	07/20/23	-1247.27	-1212.73	7.083573	7.087727
1Y	10/20/23	-1626.82	-1586.94	7.045618	7.050306
1.5M	01/22/24	-1947.15	-1890.02	7.013385	7.019998
1.8M	04/22/24	-2257.25	-2183.43	6.982575	6.990657
2Y	10/21/24	-2660.39	-2619.61	6.942261	6.947039
3Y	10/20/25	-3440.46	-3340.16	6.864254	6.874984
4Y	10/20/26	-4136.98	-3847.86	6.794602	6.824214
5Y	10/20/27	-5278.51	-4767.91	6.680449	6.732209
6Y	10/20/28	-6358.76	-5590.68	6.572424	6.649932
7Y	10/22/29	-7393.80	-6494.60	6.468920	6.539546
8Y	10/21/30	-8392.64	-7659.01	6.349036	6.443099
9Y	10/20/31	-9797.52	-8837.75	6.228548	6.325255
10Y	10/20/32	-11029.23	-10048.52	6.105377	6.204148
15Y	10/20/37	-16983.20	-15861.72	5.509980	5.622828
20Y	10/20/42	-21324.85	-20090.22	5.075815	5.199978
25Y	10/21/47	-23731.92	-22361.77	4.835108	4.972823
30Y	10/21/52	-25279.86	-23773.98	4.680314	4.831602

Forward

$$F = S \times \frac{1 + i_d}{1 + i_f}$$

Where:

- F = Forward exchange rate
- S = Current Spot exchange rate
- i_d = Interest rate of the domestic country
- i_f = Interest rate of the foreign country

Interest rate parity

- Say you can invest in either domestically or abroad, what are your options?
- Option A
 - Invest in the domestic asset – no FX involved
 - Result after one period: $(1 + i_d)$
- Option B
 - Exchange for 1 unit of domestic currency into foreign currency at S today
 - You grow this amount over time (you need to “switch the currency” so either $1/S$ or $1/F$)
 - $1/S * (1 + i_f)$
 - You swap back the return into domestic currency via the forward rate
 - $1/S * (1 + i_f) * F$
- Option A and Option B must have the same return (for there to be “parity”):

$$(1 + i_d) = \frac{1}{S} \times (1 + i_f) \times F$$

Forward

$$F = S \times \frac{1 + i_d}{1 + i_f}$$

Where:

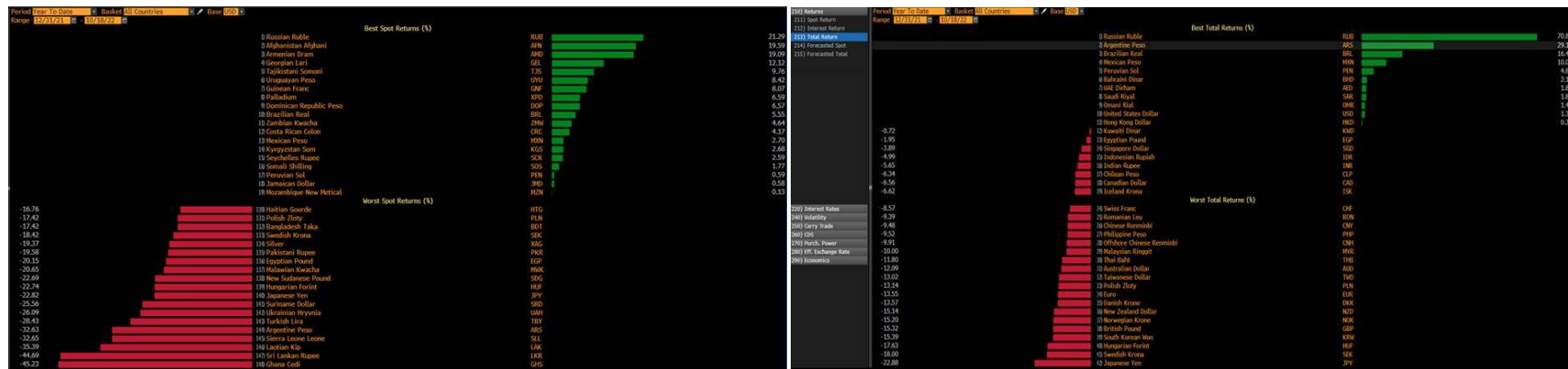
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Interest rate parity

- If the equation doesn't hold – there's opportunity for arbitrage – long one currency and short the other
- Usually, it holds because of arbitrage via carry trade

Carry Trades

- If one believes that interest rate parity will not materialise, one can simply borrow the low-interest rate currency at that country's interest rate and buy the higher interest rate currency to earn its higher rate
- Effectively you earn the interest rate differential



Correlations

- USD, GBP/EUR – risk-on/risk-off
- CHF – gold
- NOK and CAD – oil
- AUD and NZD – copper
- All currencies – government bond yields

Technical analysis

- Technicals are widely used when trading FX, even in professional settings such as banks, HFs
- Essentially, they're a way of assessing market sentiments and psychological levels/thresholds



Technical



Trade formulation

1. What's the consensus “story” – one-liner e.g. you see FT talking about “BOJ will hike rates”
2. Pull up the chart of the currency pair – what does the market tell you? Does the market agree with the consensus – could it reverse?
3. Evaluate fundamentals of the two countries – is one growing whilst the other is lagging, is one central bank hiking whilst another is easing? Can these fundamentals change?
4. Evaluate other relevant assets to identify correlations e.g. if copper is rising, could AUD and NZD also go up?